OMNIGUARD AI

Intelligent Pre-Publication Brand Governance & Compliance Platform

Team Name: Team Layer 7 | Problem Statement: Inkloom AI Submission: CodeNexus Hackathon — Round 2 Implementation Overview

1. THE PROBLEM WE SOLVED

Marketing teams and creative agencies produce content faster than ever using generative AI and digital design tools. However, publishing content without a centralized review guardrail causes serious brand safety and legal problems:

- **Off-Palette Visuals:** Unapproved colors and outdated logos make marketing campaigns look inconsistent and unprofessional.
- **Unreadable Contrast:** Low-contrast text on image backgrounds violates web accessibility standards (WCAG) and makes ads hard to read.
- **Unsubstantiated Claims:** Overpromising phrases like "*guarantees 100% compliance*" or "*unrivaled market leader*" create severe legal liabilities and false advertising fines.
- **Manual Review Bottlenecks:** Passing drafts back and forth between designers, legal teams, and brand managers takes 3 to 5 days, slowing down marketing launches.

2. WHAT WE MADE (THE OMNIGUARD SOLUTION)

OmniGuard AI is an automated brand safety firewall for outbound marketing. Before any ad copy or creative image is published, OmniGuard AI scans it in under a second against official brand guidelines. Instead of just acting as a blocker that says "no", OmniGuard AI explains why an issue was flagged and automatically generates **one-click smart fixes** that preserve the creator's original marketing message.

WHO USES OMNIGUARD AI & WHY IT MATTERS

User Persona	Their Biggest Challenge	How OmniGuard AI Solves It
Content Creators & Marketers	Waiting days for approvals and getting vague feedback on drafts.	Instant line-by-line checks with 1-click fixes to make drafts compliant in seconds.
Brand & Design Leads	Ensuring consistent colors, fonts, and messaging across all channels.	Automated color matching and tone checks against approved master brand manuals.
Legal & Compliance Officers	Need auditable proof that every claim is verified before release.	Immutable approval ledger, multi-role sign-offs, and verifiable digital certificates.

THE 5-STAGE LIVE PRODUCT WORKFLOW

Pipeline Stage	What the System Does	What the User Sees and Does
1. Ingest	Stores and indexes company brand guidelines and fact sheets.	Upload company guideline PDFs, policy manuals, or claim sheets with one click.
2. Multi-Audit	Scans text copy and visual creative images simultaneously.	Paste copy and upload an image; the scanner checks tone, claims, color, and contrast.
3. Smart Fix	Generates pinpoint text fixes that keep the original meaning.	Review side-by-side suggestions and click "Apply All Fixes" to update copy instantly.
4. Governance	Manages department sign-offs (Brand, Legal, Security).	Authorized team members review and approve audits, generating signed certificates.
5. Analytics	Monitors policy violations and recurring mistakes over time.	View visual charts showing which channels have the most issues and where rules are unclear.

3. WHAT WE BUILT IN ROUND 2 (DELIVERED CAPABILITIES)

In Round 2, we took our initial prototype and built a complete, production-ready enterprise brand governance platform. Below are the core capabilities fully working in the live platform today:

1. Real-Time Multi-Modal Scanner (Text Copy + Visual Creatives)

Instead of checking only text, OmniGuard AI inspects draft copy and visual images in parallel. It checks marketing text for overhyped buzzwords and false guarantees, while simultaneously examining the creative image to ensure the visual elements match company standards.

2. Automated Claim & Tone Detection with Clause Citations

When an unverified claim is detected (such as "*100% compliance guaranteed*"), OmniGuard AI doesn't just flag an error—it cites the exact rule from the company brand manual (e.g., *Clause 3.2: Unsupported Absolute Guarantee*). This gives creators transparent, clear explanations for why changes are needed.

3. Intent-Preserving Smart Micro-Patches (One-Click Fixes)

Other tools regenerate the entire document or wipe out human creativity. OmniGuard AI creates surgical, targeted replacements (such as changing "*guarantees 100% compliance*" to "*helps achieve up to 94% compliance across enterprise workflows**"). Clicking "Apply All Fixes" replaces the text in real time and immediately updates the compliance score.

4. Exact Color Matching & Text Readability Checking

Using computer vision, the platform inspects uploaded ad graphics:
• **Brand Color Accuracy:** Extracts dominant colors from the image and confirms they match official brand hex codes.
• **Text Contrast & Accessibility:** Verifies that foreground text is clearly readable against background colors using modern web accessibility guidelines (APCA / WCAG 3.0).

5. Brand Guidelines Knowledge Base (18 Pre-Indexed Manuals)

Includes a searchable database of 18 pre-loaded enterprise guidelines covering social media, press releases, technical ads, email marketing, and legal disclosures. Teams can also upload and index custom brand manuals (.pdf, .txt, .md) directly into the platform.

6. Dedicated Pages & Smooth Navigation for Every Tab

Every platform module has its own dedicated web address (`/scanner`, `/guidelines`, `/governance`, `/analytics`, `/graph`). Users can bookmark pages, use browser Back and Forward buttons, and switch between views with fluid directional animations without losing their current draft.

7. Interactive A/B Comparison & Quick Keyboard Shortcuts

Creators can open a side-by-side comparison modal to see their original draft right next to the compliant version. Power users can press `Ctrl+K` or `Cmd+K` to open the command palette and run audits or switch views in a split second.

4. TEAM REVIEW, PROVENANCE & SYSTEM ARCHITECTURE

In a real enterprise, no single person approves an ad alone. OmniGuard AI includes a dedicated review hub that coordinates sign-offs across different departments and creates proof of compliance.

MULTI-ROLE GOVERNANCE & DEPARTMENT SIGN-OFFS

Department Role	What They Are Responsible For	Their Sign-Off Action in OmniGuard AI
Brand Director	Visual styling, logo placement, color fidelity, and tone.	Approves brand consistency or requests creative revision.
General Counsel	Legal claims, substantiation fact sheets, and disclaimers.	Approves legal claims, rejects false superlatives, or escalates risks.
Security Architect	Data privacy, AI disclosures, and provenance verification.	Confirms AI usage disclosure and verifies cryptographic integrity.
Content Creator	Initial draft creation, channel formatting, and patch review.	Submits drafts for review and accepts 1-click micro-patches.

CRYPTOGRAPHIC AI PROVENANCE CERTIFICATES

Once an asset is approved by the required departments, OmniGuard AI generates an official, tamper-proof **Certificate of Compliance**:

- **Digital Proof:** The certificate locks together the final copy, approved image colors, cited rules, and reviewer signatures into a unique digital fingerprint.
- **Instant Verification:** Anyone can take the certificate's verification code and paste it into the on-platform verifier to confirm that the asset was officially signed off and hasn't been altered.
- **Easy Export:** Certificates can be viewed and printed directly from the browser or exported as structured data.

5. HOW THE PLATFORM IS BUILT (SIMPLE ARCHITECTURE)

We built OmniGuard AI using modern, reliable technologies so that it runs fast, stays responsive, and never loses creator work:

Component	Technology Used	What It Does in Simple Terms
Web Frontend	Next.js 16, React, Tailwind CSS	Powers the responsive web interface with dedicated tab URLs, live score updates, and smooth button hover animations.
Backend Engine	Python FastAPI, Uvicorn	A high-speed server that handles copy checking, image color analysis, patch generation, and certificate verification in under a third of a second.
Database Ledger	SQLite (15 Organized Tables)	Safely stores 18 brand manuals, historical audit records, department sign-off logs, and certificate records without needing complex cloud setups.
Computer Vision	OpenCV	Accurately analyzes colors and contrast in uploaded graphics to make sure text is readable and colors match brand palettes.
Offline Fallback	Client-Side Rules Engine	Allows creators to keep scanning copy and applying fixes even if the network slows down or disconnects.

6. SYSTEM TESTING & VERIFICATION RESULTS

To ensure the platform is reliable and bug-free, we ran an automated test suite covering every feature, API endpoint, and database table:

- **FINAL AUDIT REPORT: 47 OUT OF 47 TESTS PASSED (100% SUCCESS RATE)**
- **Backend API Endpoints:** All 13 core endpoints tested and working (Tone check, Claim check, Image color check, Patch application, Sign-offs, Certificate verification).
- **Database Integrity:** All 15 database tables verified and loaded with genuine brand guidelines, audit history, and role sign-offs.
- **Image Color & Readability:** Verified that brand colors within tolerance pass, off-palette colors are flagged, and readable contrast meets standards.
- **Speed & Performance:** Copy checks run in under 20 milliseconds; image vision checks complete in under 350 milliseconds.
- **Live Navigation:** All dedicated page routes (/scanner, /guidelines, /governance, /analytics, /graph) verified active and responding with HTTP 200.

7. TEAM LAYER 7 — WHO BUILT WHAT

Team Member	Role	Key Responsibilities & Deliverables
Swastik Mittal	Data & Systems Engineer	Built the image color and readability inspection engine, designed the 15-table database structure, and organized the Brand Knowledge Base.
Aprajita	Frontend Lead	Designed the responsive web interface, created dedicated web page routes, added fluid page animations, and built the legal sign-off workflow.
Aditya Baliyan	Backend Lead	Built the high-speed FastAPI endpoints, created the multi-role review and sign-off system, implemented certificate generation, and wrote the test suite.
Nikita Singh	Full Stack Integrator	Connected the frontend to the backend APIs, built the offline fallback engine, added the command palette (Ctrl+K), and created the A/B comparison modal.

8. WHAT WE WILL BUILD IN ROUND 3 (NEXT STEPS)

If selected for Round 3, we will add three major capabilities that build directly on our working system:

Feature	What It Does	How It Helps the User
1. Bulk Campaign Ingestion	Allows uploading a ZIP folder or spreadsheet containing 50+ ads at once to scan all of them in parallel.	Marketers can audit an entire multi-channel ad campaign in a single click instead of testing assets one at a time.
2. On-Image Highlight Boxes	Draws interactive visual boxes and pin markers directly on the uploaded image showing where colors or contrast failed.	Designers see the exact spot on their graphic that needs to be fixed without any guesswork.
3. Policy Conflict Detector	Automatically detects if two brand guidelines contradict each other and provides a side-by-side merger tool.	Brand managers easily resolve conflicting rules across different regions or departments in one clean interface.